Module 10: Epigenetics - Key Points

Learning Objectives

By the end of this module, you should be able to:

1. Explain gene regulation as control of gene activity.
2. Describe DNA methylation at an introductory level.
3. Connect chromatin structure to gene accessibility.
4. Give a careful example of environmental influence on expression.
5. Distinguish epigenetic regulation from mutation.

Topics

- Gene regulation as control
- DNA methylation
- Histone and chromatin change
- Environment and expression
- Epigenetics versus mutation

Key Terms to Know

- **Gene regulation** - Control of when and how strongly genes are used.
- **Epigenetics** - Heritable or persistent expression changes without DNA sequence change.
- **DNA methylation** - Chemical marking of DNA often associated with reduced transcription.
- **Histone** - Protein around which DNA is packaged.
- **Chromatin** - DNA-protein complex that can be open or compact.
- **Transcription factor** - Protein that helps control transcription.
- **Mutation** - Change in DNA sequence.
- **Expression** - Use of gene information to make RNA or protein.

Core Contents

1. Cells regulate which genes are active rather than using every gene equally.
2. DNA methylation can reduce transcription without changing base sequence.
3. Histone and chromatin changes alter access to DNA.
4. Environmental conditions can influence gene expression patterns.
5. Epigenetic change affects gene activity, while mutation changes DNA sequence.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the same-numbered lab as the evidence check for the module.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-10_epigenetics.md`

Generated Visuals

- [Module 10: Epigenetics Evidence Map]
- [Module 10: Epigenetics Reasoning Sequence]
- [Module 10: Epigenetics Retrieval and Lab Check]